

Mastering Immutable Infrastructure: A Step-by-Step Guide to Building Golden AMI

You can leverage Packer, Ansible, and Testinfra to automate the creation and testing of golden AMIs, ensuring your infrastructure is configured correctly. Here's how!

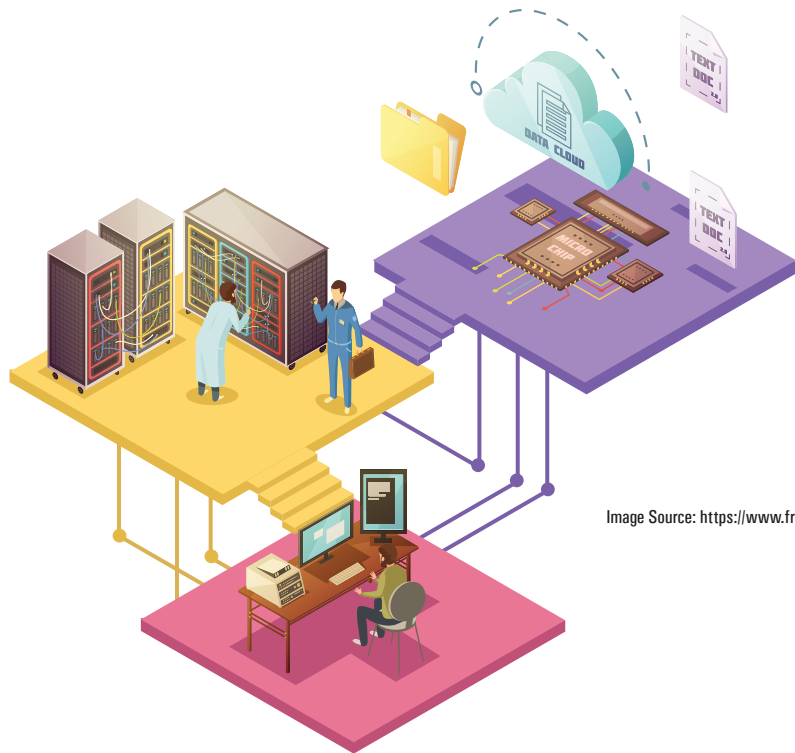


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This article covers a step-by-step process of building a golden AMI using Packer, Ansible and TestInfra. It also demonstrates how to integrate these tools to master immutable infrastructure.

In recent years, immutable infrastructure has become the most popular approach to deploying applications in the cloud. It enables developers to create secure, scalable, and repeatable infrastructure. Immutable infrastructure involves creating machine images, also known as golden AMIs, that are pre-configured and ready to run applications. These machine images are then deployed as-is and never modified, ensuring reliable and consistent infrastructure.

Before we begin, let's quickly understand the basic terminologies and get familiar with the tools we are going to use.

- **Immutable infrastructure:** This approach to infrastructure management emphasises creating and deploying infrastructure in a non-changing state. In other words, an

immutable infrastructure cannot be modified once created. Instead of modifying the existing infrastructure, a new version is created every time changes are made. Immutable infrastructure has several benefits, such as easier testing, more efficient deployments, and better security.

- **Golden AMI:** It is an Amazon machine image containing the operating system, configuration, and applications required to run an application or service. It is the foundation for creating instances in AWS. Ensuring that the AMI is up-to-date, secure, and consistent is crucial.
- **Packer:** This tool creates identical machine images for multiple platforms from a single source configuration. It allows you to automate the process of building machine images, which can be used for deployment, testing, and development. Packer supports multiple builders, including Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure.